Synopsis

Air Quality Prediction using Machine Learning

Problem Id: 15

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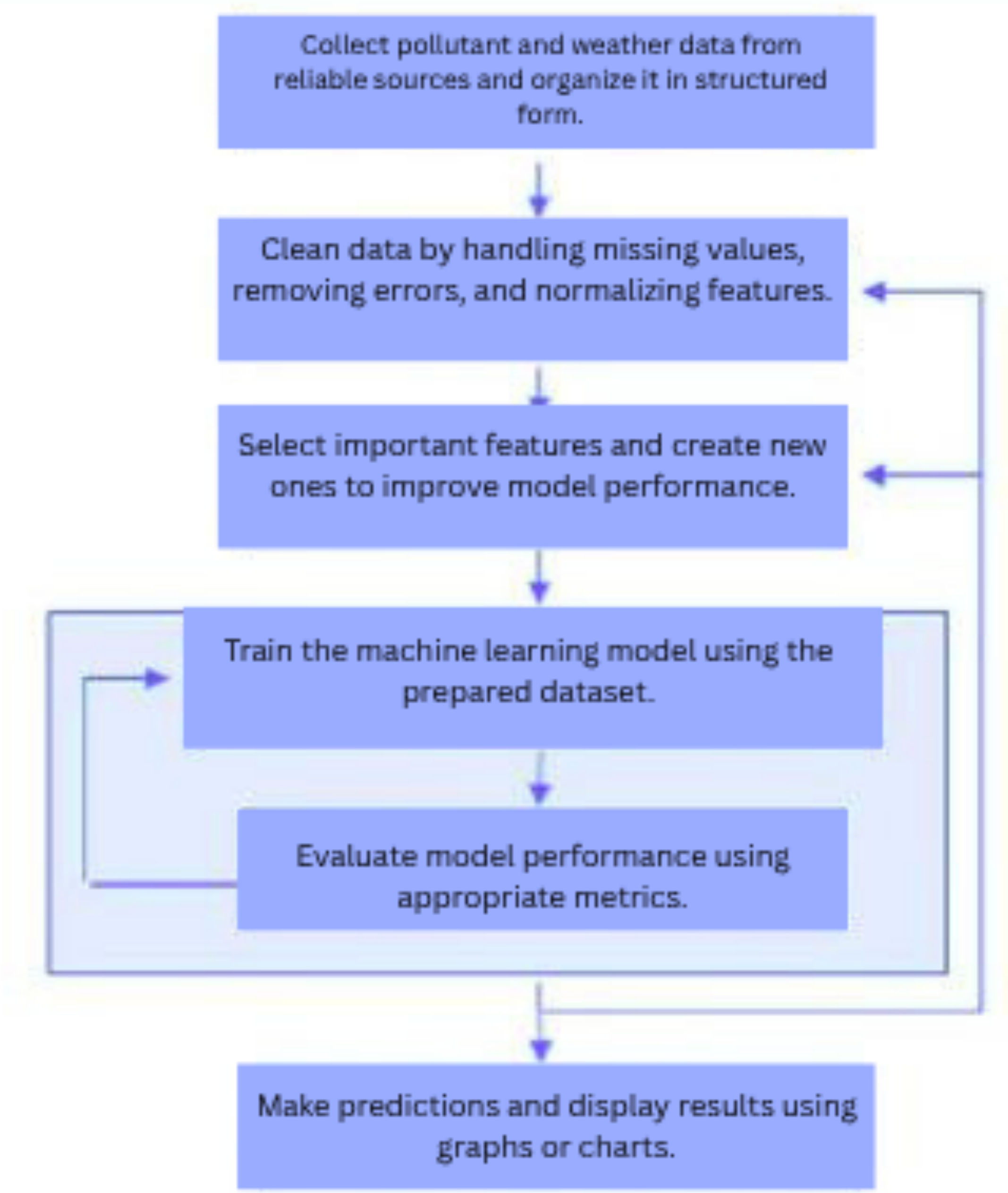
- **Problem Description:**

Air pollution is a growing environmental and public health concern worldwide. High concentrations of pollutants such as PM2.5, PM10, carbon monoxide (CO), nitrogen dioxide (NO_2), and sulfur dioxide (SO_2) can lead to serious health issues including respiratory diseases, cardiovascular problems, and reduced life expectancy. Therefore, accurate prediction of air quality is essential for taking timely preventive measures.

This project aims to develop a machine learning-based system to predict air quality levels using historical pollutant data and meteorological parameters like temperature, humidity, and wind speed. By analyzing patterns and relationships within the data, the model can forecast future air quality conditions.

In real-life applications, such predictive systems are used by environmental agencies and smart city infrastructures to issue early warnings, manage pollution sources, and safeguard public health. The project highlights the importance of data-driven decision-making in addressing environmental challenges and improving quality of life.

- **Steps of Implementation:**



- **Expected modules/ libraries to be used:**

- **Evaluation Metrics:**

  - **Mean Absolute Error (MAE)** – Measures the average difference between actual and predicted values.
  - **Root Mean Squared Error (RMSE)** – Measures error by giving higher weight to large differences.
  - **R2 Score (Coefficient of Determination)** – Indicates how well the model explains the data.

Plots:

- **Line Plot** – To compare actual vs predicted values over time.
- **Scatter Plot** – To visualize the relationship between actual and predicted values.
- **Histogram** – To understand distribution of dataset values.

- **Required topics from the subject:**

Strings:
Strings are used for handling textual data such as column names and categorical values like wind direction. They also help in formatting and cleaning inconsistent data entries.

Files:
File handling is used to read input datasets (CSV files) and store processed data or results. It allows the program to work with large datasets efficiently by reading and writing data from external storage.

Arrays:
Arrays are used to store and process multiple values efficiently. In data analysis, they help in handling large sets of numerical data.

Data Structures:
Data structures like lists, sets, and dictionaries are used to store and organize data efficiently for processing and analysis.

Data Preprocessing:
This involves cleaning the dataset by handling missing values, removing outliers, and transforming data into a suitable format for model training.

- **Platform Required:**

Jupyter Notebook

- **Books and Link Sources:**

Hands-On Machine Learning with Scikit-Learn, Keres & TensorFlow – Aurélien Géron
Python for Data Analysis – Wes McKinney
Scikit-learn Official Documentation – https://scikit-learn.org
Kaggle Dataset Platform – https://www.kaggle.com
UCI Machine Learning Repository – https://archive.ics.uci.edu